

SpaceX Falcon 9

First Stage Landing Prediction



IBM Applied Data Science Capstone

Predicting whether the Falcon 9 first stage will land successfully using launch, payload, orbit, site and booster attributes.

GitHub repository: <https://github.com/MCerro/IBM-Data-Science-Capstone-SpaceX>

Executive Summary

90

Falcon 9 launch records analysed

66.7%

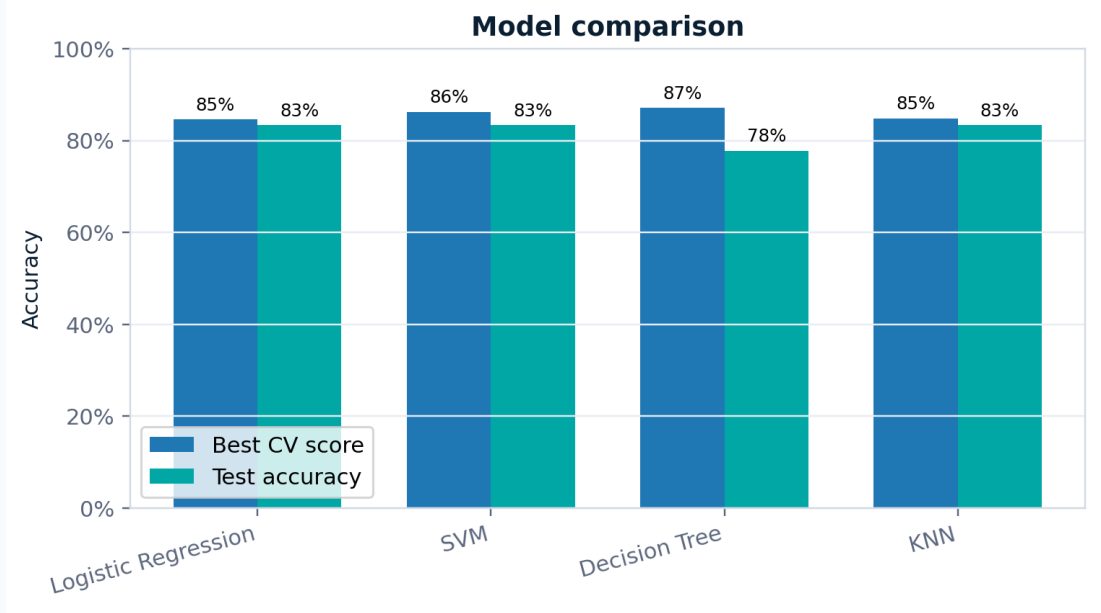
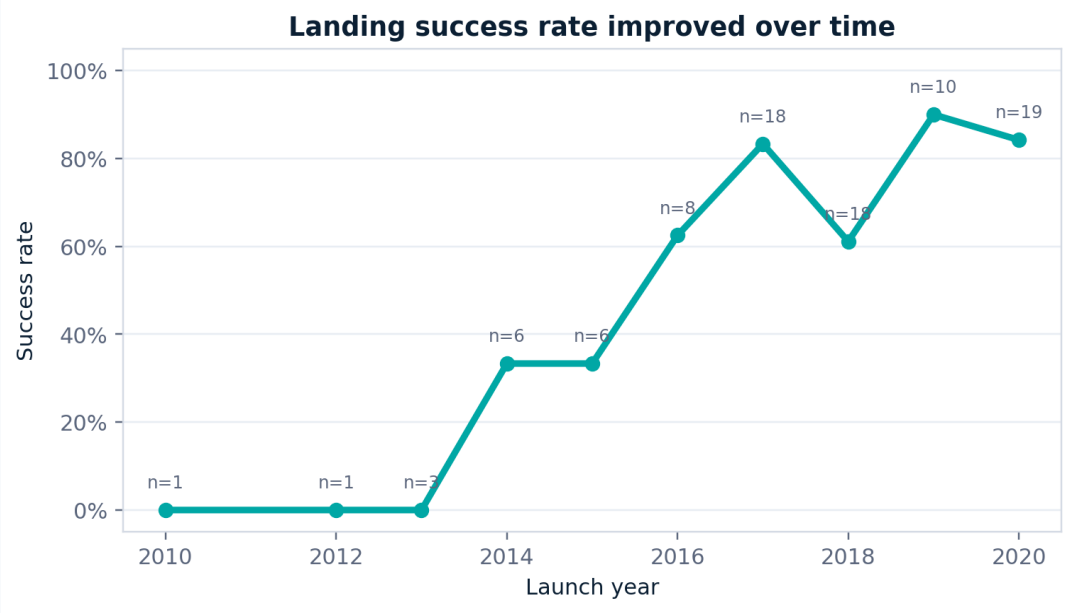
Overall landing success rate

83

Machine-learning features after encoding

83.3%

Best held-out test accuracy



Main finding: the landing outcome is predictable from the engineered launch dataset, but model separation is limited by small sample size and class imbalance in older missions.

Results calculated from dataset_part_2.csv and dataset_part_3.csv generated for this repository.

Project Objective and Analytical Questions

Objective

Build a reproducible data-science workflow to analyse Falcon 9 launches and predict first-stage landing success.



- Can the first stage landing outcome be predicted?
- Which launch site has the highest success rate?
- How do payload mass, orbit and flight number relate to outcome?
- What does the Folium site map reveal about launch-site location context?
- Which classification model performs best on the test set?

- Unit of analysis: individual Falcon 9 launch
- Target variable: Class = 1 for successful landing/recovery, 0 otherwise
- Prediction task: binary classification
- Primary evaluation: accuracy on held-out test data plus confusion matrices

Data Sources and Repository Artifacts

Artifact	Role in project	Rows × Columns
dataset_part_1.csv	Clean API-derived launch data	90 × 17
dataset_part_2.csv	Wrangled data with landing Class label	90 × 18
dataset_part_3.csv	Encoded ML feature matrix	90 × 83
SpaceX.csv	SQL laboratory dataset	101 × 10
spacex_web_scraped.csv	Wikipedia snapshot extraction	121 × 11
spacex_launch_geo.csv	Launch-site location data for Folium	56 × 13
spacex_launch_dash.csv	Plotly Dash input dataset	56 × 7

Data provenance rule

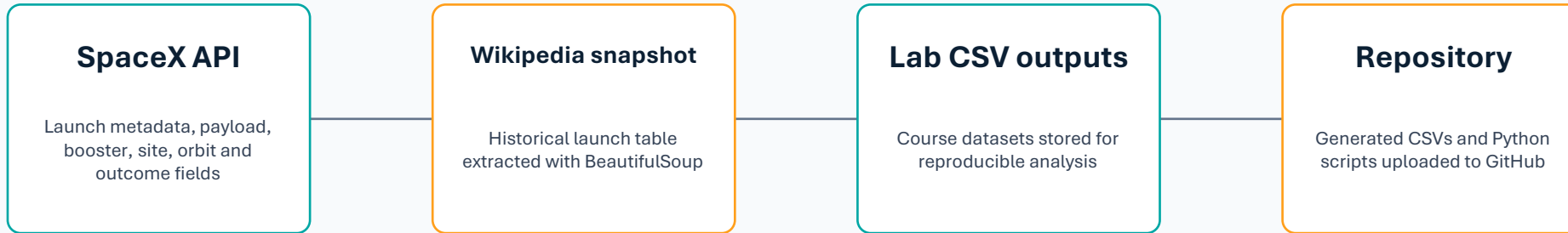
Only observed files and derived outputs are used in this deck; missing notebooks are not assumed.

GitHub repository contents

- CSV datasets
- Data-generation script
- Dash dashboard script
- README.md
- PowerPoint and PDF after final export

The final PDF includes the confirmed GitHub repository link: <https://github.com/MCerro/IBM-Data-Science-Capstone-SpaceX>

Data Collection Methodology



Implementation evidence

- generate_spacex_data.py downloads or consumes source CSVs and derives downstream datasets.
- dataset_part_2.csv is derived from Outcome using the landing-success Class label.
- dataset_part_3.csv is produced by one-hot encoding categorical launch and booster features.

Reproducibility note

- The Python script validates final row/column shapes before accepting outputs.
- The web-scraped dataset uses the fixed IBM Wikipedia snapshot and keeps the original lab schema.
- No additional rows or performance metrics are fabricated.

Data Cleaning and Transformation

Wrangling operations

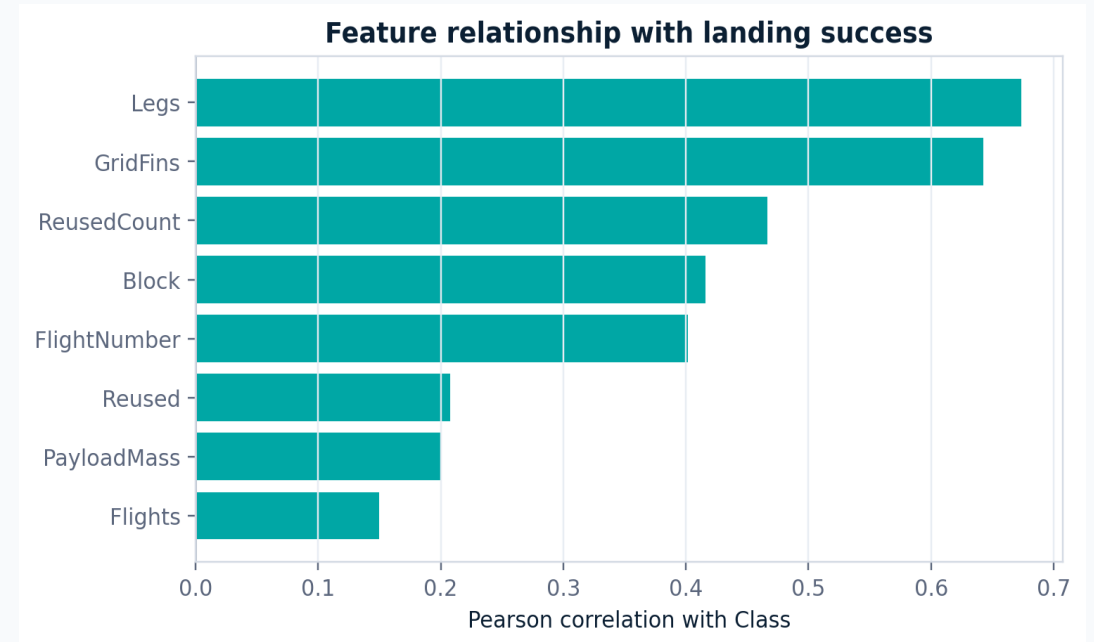
- Built the binary target: Class = 1 if the booster landing outcome indicates successful landing/recovery.
- Retained analytical fields such as FlightNumber, PayloadMass, Orbit, LaunchSite, Flights, GridFins, Reused, Legs, LandingPad, Block, ReusedCount and Serial.
- Encoded categorical variables with one-hot encoding for classification models.
- Standardized features before model training.

17 → 18

Columns after Class label creation

18 → 83

Feature expansion after one-hot encoding



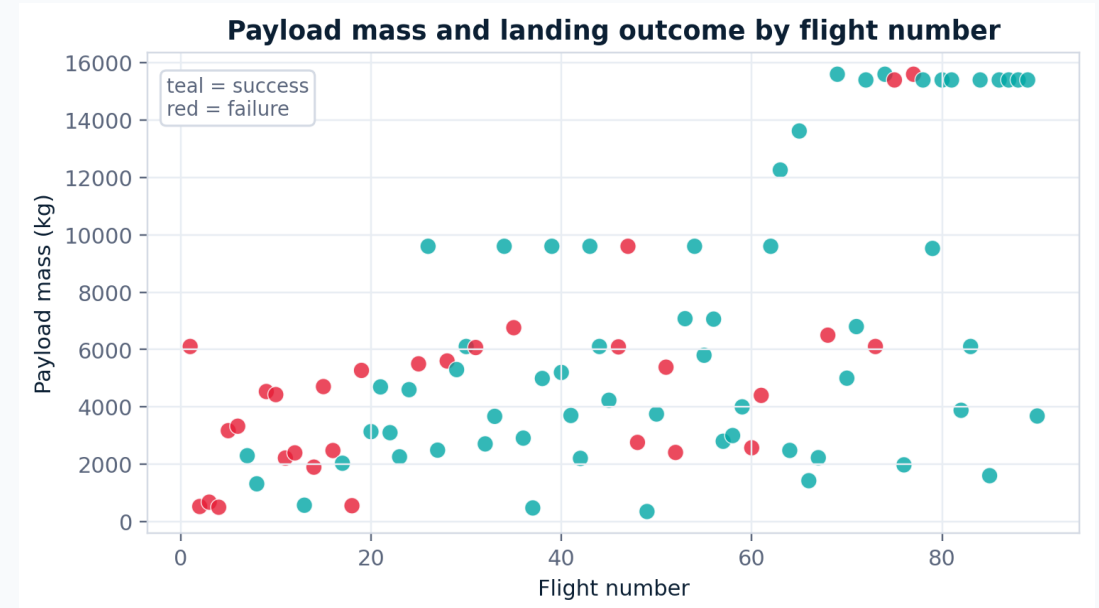
Variables with the strongest simple relationship to Class include flight number, grid fins, legs and reuse-related fields. Correlation is descriptive, not causal.

Feature relationship chart uses Pearson correlation against the binary Class target.

EDA and Interactive Analysis Methodology

Exploratory workflow

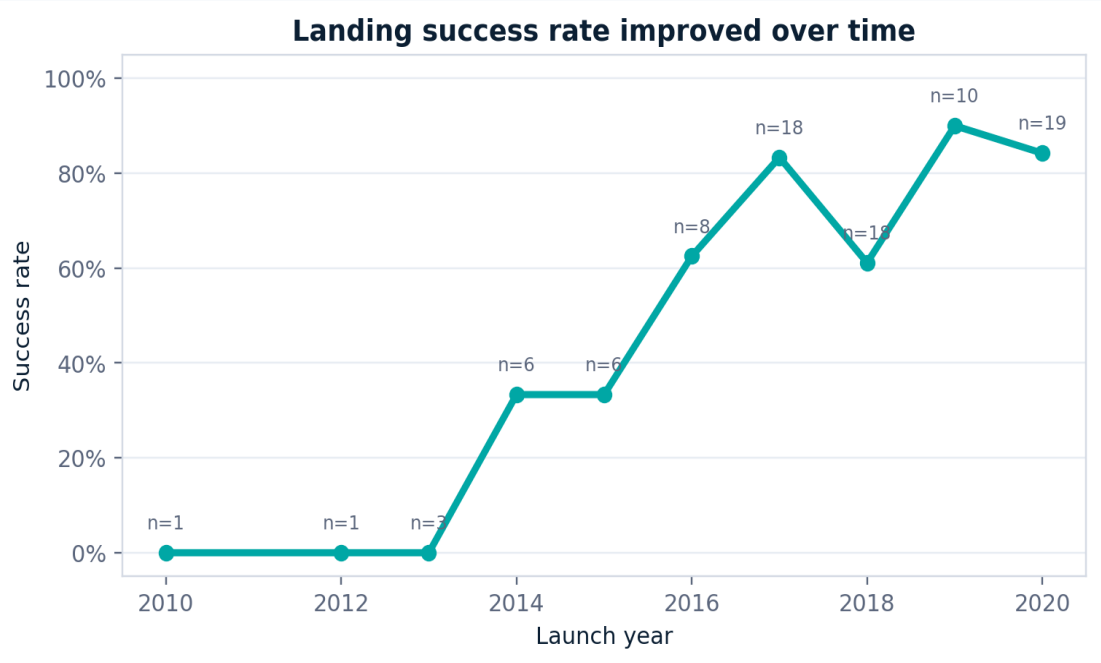
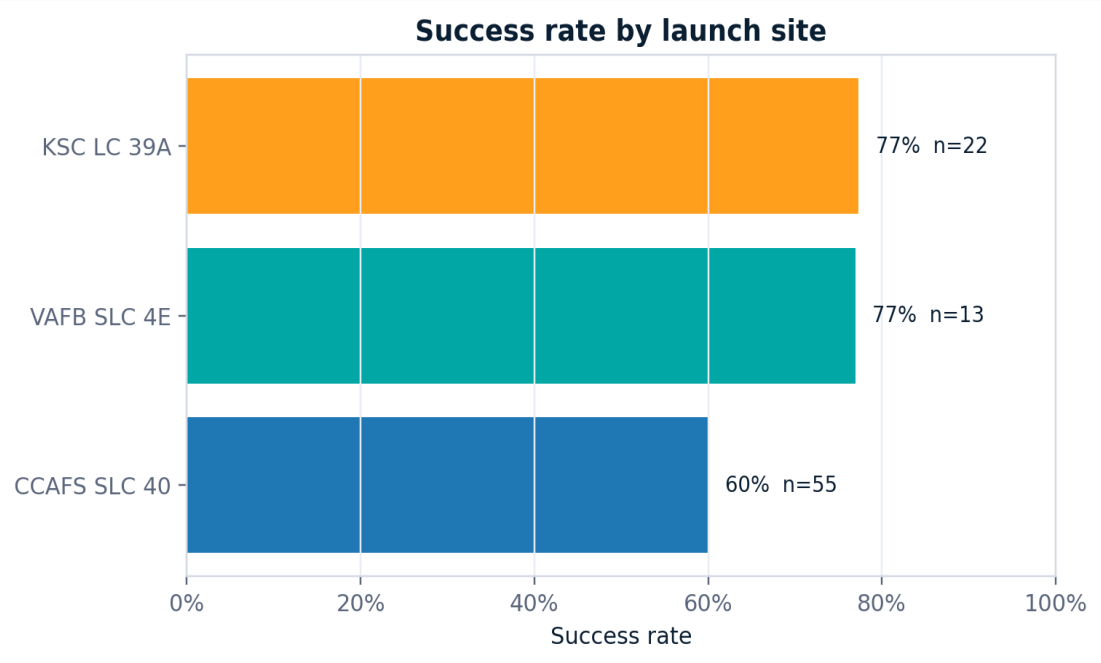
- Distribution checks for launch sites, orbit types, payload mass and landing outcomes.
- Group-by summaries for success rate by site, orbit, year and payload band.
- Scatter plots to inspect payload and flight number patterns.
- SQL-style aggregation for totals, payload averages and outcome breakdowns.
- Folium map to evaluate launch-site geography and marker-level context.
- Plotly Dash dashboard for interactive filtering by site and payload range.



EDA question mapping

Site → launch location performance · Orbit → mission profile · Payload
→ vehicle demand · FlightNumber/Year → learning curve over time

EDA Results: Launch Site and Time Trend



KSC LC 39A

Highest site success rate: 77.3% across n=22

Improving

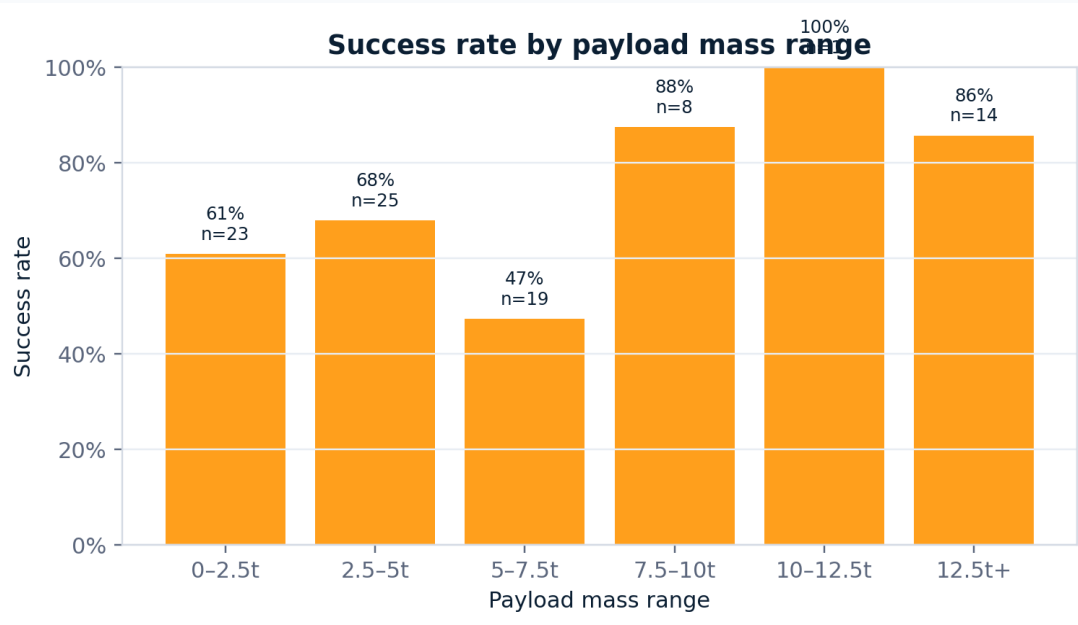
Success rate rises noticeably in later flight years

60:30

Successful vs unsuccessful landings in dataset_part_2

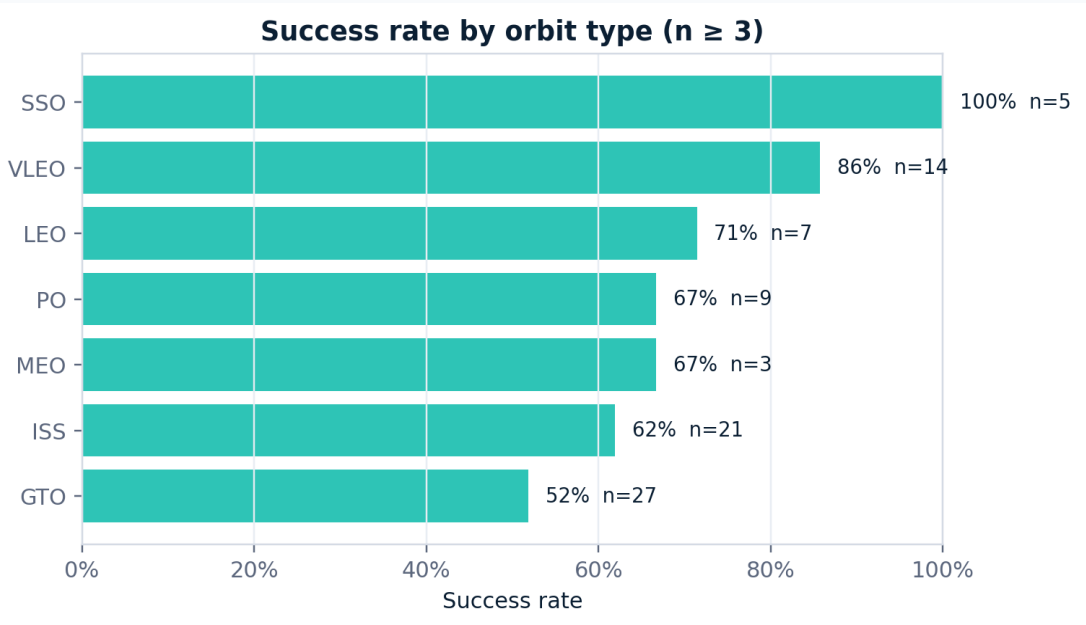
Site results are based on dataset_part_2.csv, not on undocumented external data.

EDA Results: Payload and Orbit Effects



Payload insight

Payload mass alone does not define outcome; success varies by payload range and overlaps strongly between successful and failed missions.



Orbit insight

The best-performing orbit categories in this dataset have high success rates, but small category counts mean results must be interpreted carefully.

Orbit chart only shows categories with at least three launches.

EDA Results with SQL-Style Aggregations

Representative queries

```
SELECT Launch_Site, COUNT(*) AS Launches
FROM SpaceX
GROUP BY Launch_Site
ORDER BY Launches DESC;
```

```
SELECT Orbit, AVG(PAYLOAD_MASS__KG_)
FROM SpaceX
GROUP BY Orbit
ORDER BY AVG(...) DESC;
```

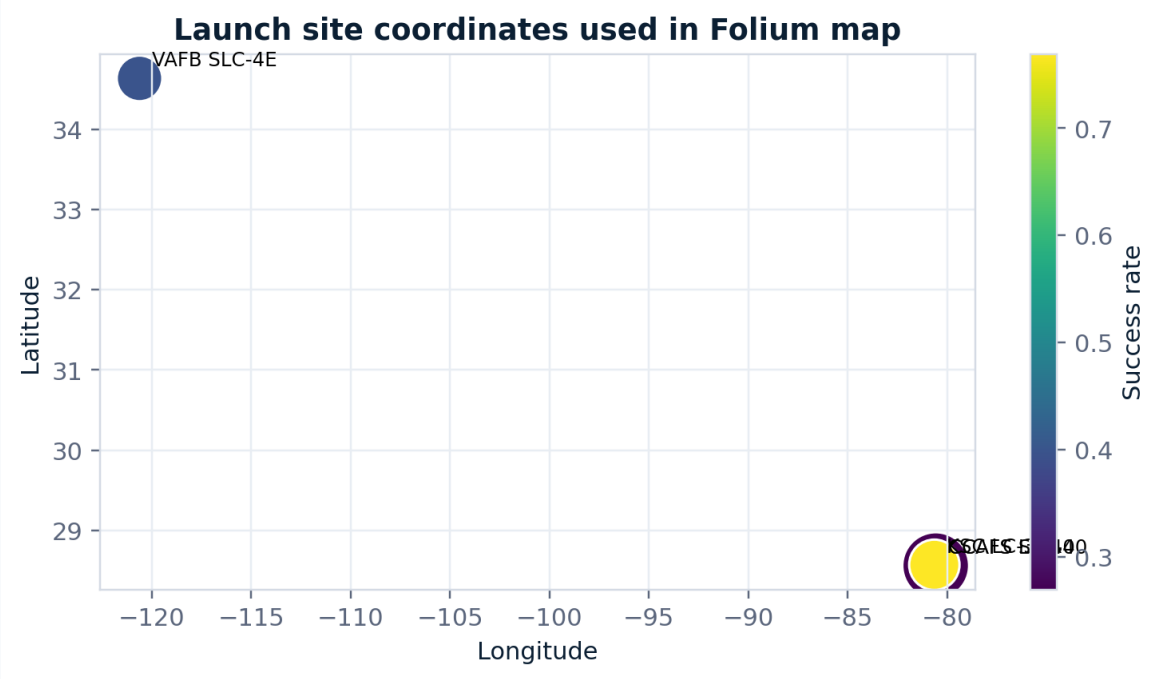
SQL output confirms launch distribution by site and highlights that payload mass differs by mission orbit. These results support the feature set used for predictive modelling.

Launch site	Launches
CCAFS SLC-40	34
CCAFS LC-40	26
KSC LC-39A	25
VAFB SLC-4E	16

Orbit	Avg payload kg
Sub-orbital	12,050
LEO	11,063
Polar LEO	8,070
GTO	5,001
MEO	4,341
LEO (ISS)	3,301

Tables derived from SpaceX.csv using pandas-equivalent SQL aggregations.

Interactive Folium Map Results



Map features implemented

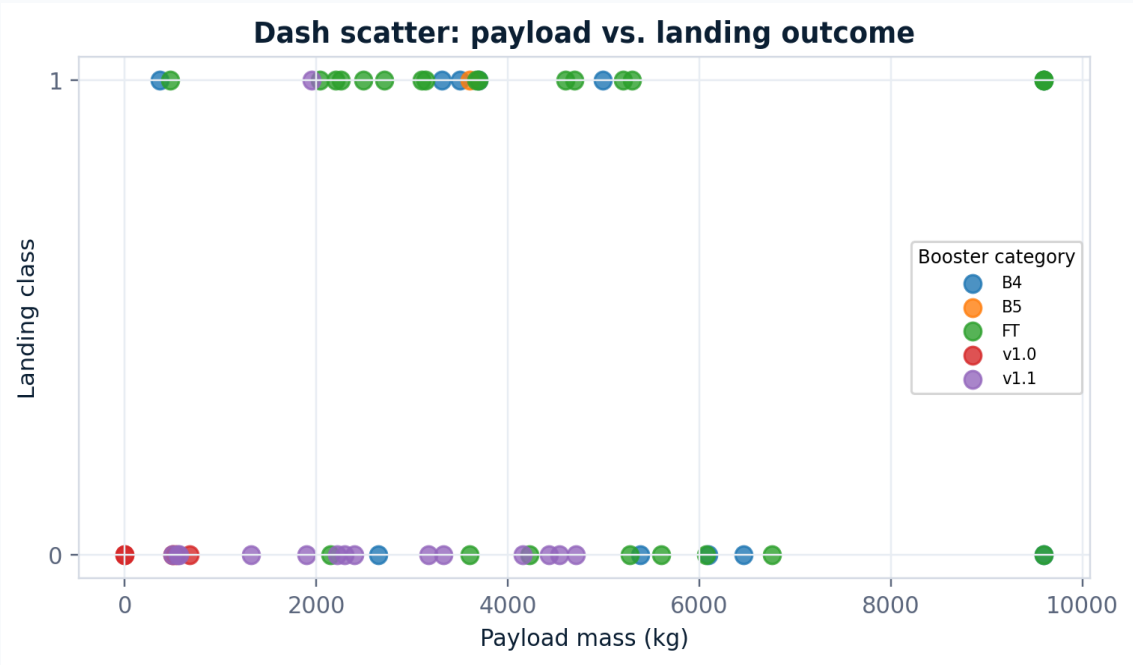
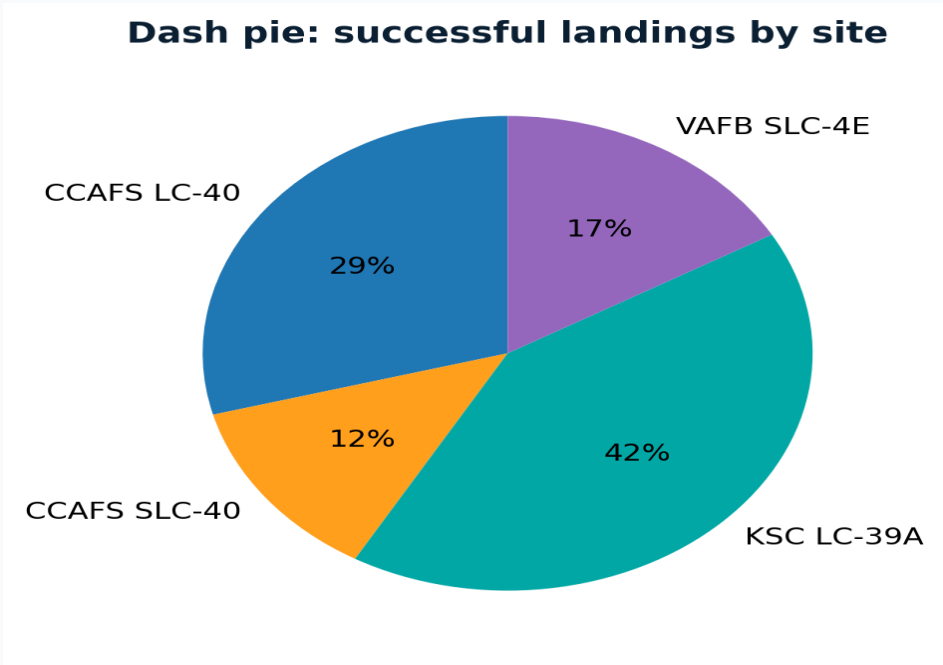
- Launch site markers using latitude/longitude from spacex_launch_geo.csv.
- Circle markers showing successful and unsuccessful landing outcomes.
- Distance-measure workflow for coastline, city, road and railway context.
- Clustered view helps compare launch-site concentration and geographic constraints.

Actual sites in dataset

Site	Lat	Long
CCAFS LC-40	28.562	-80.577
CCAFS SLC-40	28.563	-80.577
KSC LC-39A	28.573	-80.647
VAFB SLC-4E	34.633	-120.611

The static coordinate plot represents the Folium map input; the notebook contains the interactive map behaviour.

Plotly Dash Dashboard Results



Dashboard interactions

- Dropdown filters All Sites or one specific launch site.
- Payload range slider restricts visible launches.
- Pie chart compares successful landings by site or outcome split within a site.
- Scatter plot links payload mass, booster version category and landing class.

Charts mirror the dashboard logic in `spacex_dash_app.py` using `spacex_launch_dash.csv`.

Predictive Analysis Methodology

Target and feature set

- Target: Class, where 1 = successful landing/recovery and 0 = unsuccessful landing/recovery.
- Features: 83 numeric columns from dataset_part_3.csv after one-hot encoding.
- Train/test split: 80% training, 20% held-out test, random_state=2.
- Scaling: StandardScaler fitted to the feature matrix before model training.
- Model tuning: GridSearchCV with 10-fold cross-validation.

Models compared

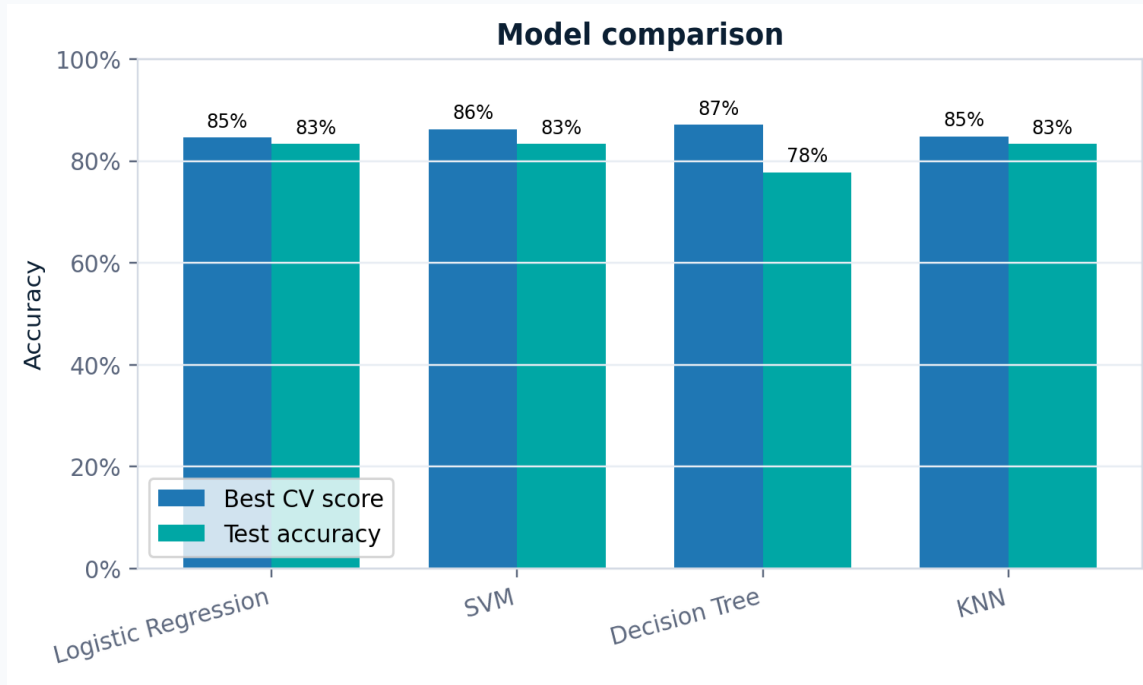
Model	Tuning focus
Logistic Regression	Regularisation C
SVM	Kernel and C
Decision Tree	Depth and split rules
KNN	k, distance metric, weighting

Evaluation principle

Use the held-out test score and confusion matrix for final comparison; use cross-validation score to judge tuning stability.

Model scores in the next slides are calculated directly from dataset_part_2.csv and dataset_part_3.csv.

Predictive Model Results



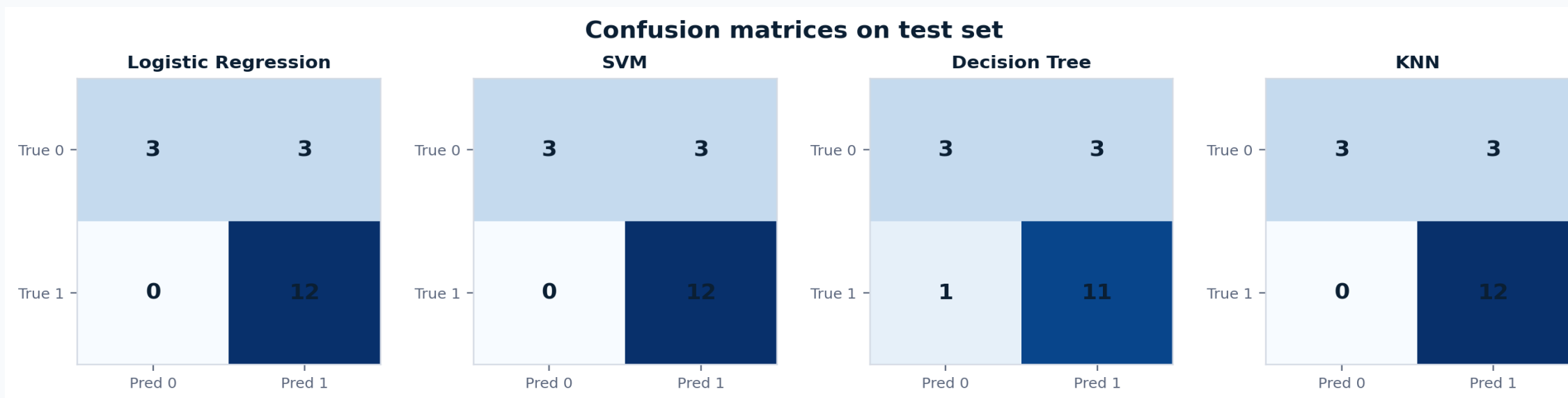
Model	Best CV	Test	Best hyperparameters
Logistic Regression	84.6%	83.3%	C=0.01, solver=lbfgs
SVM	86.3%	83.3%	C=0.01, gamma=scale, kernel=linear
Decision Tree	87.1%	77.8%	criterion=gini, max_depth=4, min_samples_leaf=4, min_samples_split=2
KNN	84.8%	83.3%	n_neighbors=10, p=1, weights=uniform

Model selection

Logistic Regression, SVM and KNN tie at 83.3% test accuracy. SVM has the highest cross-validation score among the tied models, while Decision Tree has the highest CV score overall but lower test accuracy.

Because the test set has 18 launches, one prediction changes test accuracy by 5.6 percentage points.

Confusion Matrices and Error Pattern



Interpretation

- The strongest models classify all successful landings in the test set correctly.
- Most errors are false positives: predicted success where the true class is failure.
- This pattern matters operationally because overestimating success would understate launch risk.

Recommended choice

Use SVM as the primary model in the final story because it ties for best held-out accuracy and has the highest cross-validation score among the 83.3% test-accuracy models.

Confusion matrices use label order 0 = unsuccessful, 1 = successful.

Innovative Insights, Limitations and Future Work

Insights supported by current data

- Success appears to improve with flight experience and time, consistent with an operational learning curve.
- Launch site, orbit and booster configuration all add signal beyond payload mass alone.
- Geospatial context is valuable for storytelling and safety interpretation even when it is not used directly in the classifier.

Future work

- Add newer launches and retrain the model with a time-based split.
- Report precision, recall, F1 and ROC-AUC in addition to accuracy.
- Use calibration curves to estimate probability reliability.
- Group booster serials into reusable engineering features instead of individual one-hot columns.
- Add weather, mission complexity and landing-zone environmental variables.

Limitations

- Small sample size: 90 rows after wrangling.
- High-cardinality booster serial variables may overfit.
- Accuracy alone hides false-positive risk.
- Historical outcomes may not generalise to new vehicles or mission profiles.

Submission note

GitHub repository URL confirmed and included:
<https://github.com/MCerro/IBM-Data-Science-Capstone-SpaceX>

Conclusions

Can landing success be predicted?

Yes — with this dataset, tuned classification models achieved up to 83.3% accuracy on the held-out test set.

KSC LC 39A

Best launch-site success rate: 77.3%

Payload matters

But not as a standalone predictor

SVM selected

Best CV among tied test winners

Final technical conclusion

The most defensible model story is not just “highest accuracy”: it combines EDA evidence, site/orbit/payload patterns, an interpretable data pipeline, and awareness of the model’s false-positive risk.

Next deliverable: export this PowerPoint to PDF for submission.